

Vishwanath Guruvayur

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EXPERIENCE

Senior Data Scientist, Capital One

Nov 2025 - Present

- Developing **real-time device trust ML models** for Account Takeover (ATO) fraud detection, owning feature engineering, model training, evaluation, monitoring, and automated retraining in production.
- Building **graph-based fraud detection models** by constructing behavioral graphs and learning neighborhood-level representations to identify coordinated fraud patterns and higher-order fraud rings beyond point-in-time signals.

AI Research Scientist, University of Virginia

Jun 2025 - Present

- Part-time researcher designing a **conversational AI tutor** by building a **multimodal RAG** system on classroom content, helping students discover and comprehend relevant educational material while improving accessibility of learning resources.
- Co-authored a **peer-reviewed paper** ([link](#)) presented at AI for Education 2025, introducing novel techniques for educational content retrieval and evaluation; awarded "Most Compelling Data Visualizations" at UVA's Capstone Showcase.

Data Scientist (Team Lead), Piramal Pharma Ltd.

Jan 2022 - Jun 2024

- Established and **led a new Data Science team** for the Critical Care division, embedding analytics and automation into core business workflows and mentoring junior scientists to deliver compliant, production-grade data solutions.
- Built and deployed **30+ production ML pipelines** with model governance, automated retraining, and performance monitoring using Databricks and SQL, improving overall operational efficiency by upto 85%.
- Partnered with the CFO to process terabytes of market data using Python and Spark on AWS, developing **predictive ranking models**, market segmentation, and Power BI dashboards to guide investment and portfolio strategy.
- Prevented \$1.2M in quarterly write-off losses by designing **inventory risk-ranking models** using causal inference and time-series forecasting under volatile demand conditions.

Data Science Intern, Piramal Pharma Ltd.

Jul 2021 - Dec 2021

- Drove a **15% quarterly increase** in market penetration across 400,000 retailers by building the company's first product recommendation system, coding a tree-based custom FP-Growth algorithm from scratch.
- Boosted high-performing sales reps' **retention to 93%** by developing an attrition prediction model using an ensemble of XGBoost and Logistic Regression on historical sales and HR data.

CERTIFICATIONS

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| AWS Certified Solutions Architect – Associate (SAA-C03), Amazon Web Services | May 2025 - May 2028 |
| AWS Certified Machine Learning – Specialty (MLS-C01), Amazon Web Services | Jun 2025 - Jun 2028 |

SKILLS

Analytics: Model bias detection, risk-aware ML, model governance, statistical validation, hypothesis testing

Machine Learning: Logistic Regression, GBMs, XGBoost, Ranking/Scoring, NLP, Recommendation Systems

LLMs & GenAI: LangChain, HuggingFace, Multimodal RAG, Prompt Engineering, Fine-tuning and PEFT (LoRA, IA3)

Languages: Python, R, SQL, JavaScript

Visualization: Power BI, Plotly Dash, d3.js, Matplotlib

Data Engineering and Cloud: Spark, Hadoop, Docker, Conda, APIs, Web Scraping, AWS Certified, Azure Databricks

EDUCATION

University of Virginia (UVA)

Jun 2024 - May 2025

Master of Science in Data Science

GPA: 4.0/4.0

Birla Institute of Technology and Science (BITS Pilani)

Aug 2018 - Dec 2021

B.E. Mechanical Engineering, Minor in Data Science

GPA: 8.8/10